

Mikko De Torres

Mechatronics & AI Engineer

Mechanical Design · CAD · 3D Printing · Robotics · Prototyping

[email] [phone]

LinkedIn: [link] YouTube: [link] GitHub: [link]

I am a mechatronics engineer who designs mechanical systems and builds them. My work runs from CAD modelling and 3D printing through electronics, control code and testing, then into documentation and video. I teach robotics at Batangas State University and run a YouTube channel where I explain and film engineering work.

MECHANICAL DESIGN & CAD

SolidWorks, Onshape, AutoCAD. Designed the mechanical structure and folding mechanism of a DOST-funded solar power station, built and deployed as a working prototype.

3D PRINTING & FABRICATION

Printed, fitted and assembled robot arm kits used as teaching platforms. Teach Basic Workshop and Machining, including measurement, 3D printing and BOM work.

ELECTRONICS & CODE

Arduino, ROS, C++, Python, MATLAB. Built a 3-DOF manipulator controlled through ROS and Arduino, with kinematics solved in Python.

CONTENT CREATION

Run a Mechatronics and Robotics Engineering YouTube channel. Script, film and edit technical videos, and maintain public GitHub repositories used in my lectures.

EDUCATION

BS Mechatronics Engineering, Batangas State University (2012)

MS Artificial Intelligence, Batangas State University (in progress, 24 units)

Collapsible Solar Power Station for Farms

[SolidWorks render or photo of the deployed station]

WHAT IT IS

A portable solar power station for farms, funded by DOST-TAPI. It carries photovoltaic panels and a battery unit, and folds down so it can be moved, deployed and stored easily.

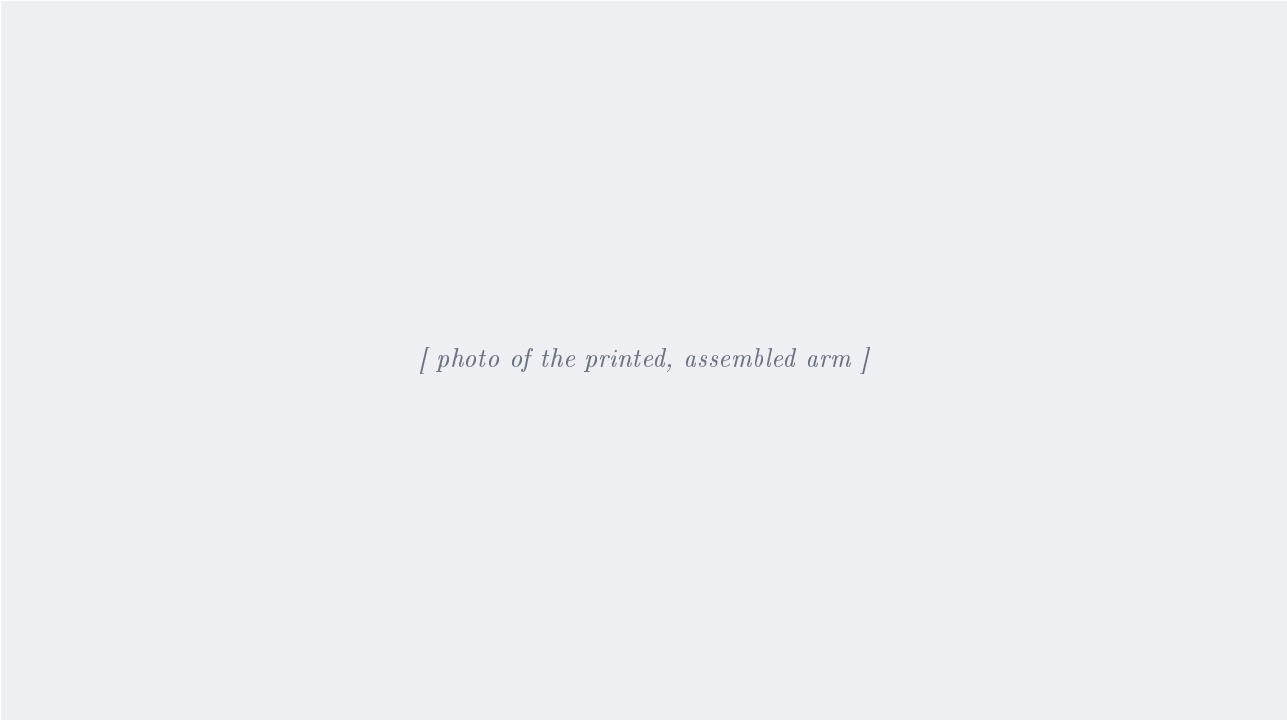
MY ROLE

Designed the mechanical structure and the collapsing mechanism in SolidWorks. Produced the drawings, the brochure and the instruction materials. The unit was fabricated and deployed in the field.

TOOLS

SolidWorks · technical documentation · fabrication and assembly

6-DOF Robot Arm Kit



[photo of the printed, assembled arm]

WHAT IT IS

A six-axis robot arm, 3D printed and assembled as a working teaching platform for my robotics classes. Students use it to see kinematics and control on real hardware.

MY ROLE

[one line — what you printed, fitted, adjusted, assembled]

TOOLS

3D printing · [CAD package] · [controller]

[photo of the built manipulator, or ROS / RViz screenshot]

WHAT IT IS

A three-axis articulated manipulator built and controlled through ROS and Arduino. This is the project where mechanical, electronics and software meet.

MY ROLE

Built the manipulator, wrote the real-time control in C++ under ROS, and solved the kinematics in Python for motion planning.

TOOLS

ROS · Arduino · C++ · Python



[sample inspection image or result plot]

WHAT IT IS

A vision system that inspects weld quality on stainless steel using HDR cameras and deep learning, built in Python.

MY ROLE

Built the full pipeline. Used transfer learning on a pre-trained VGG16 network for feature extraction, then adapted it to classify weld defects. Ran the testing and evaluation.

TOOLS

Python · HDR cameras · deep learning (VGG16, transfer learning)



[SCARA V3 simulation screenshot]

WHAT IT IS

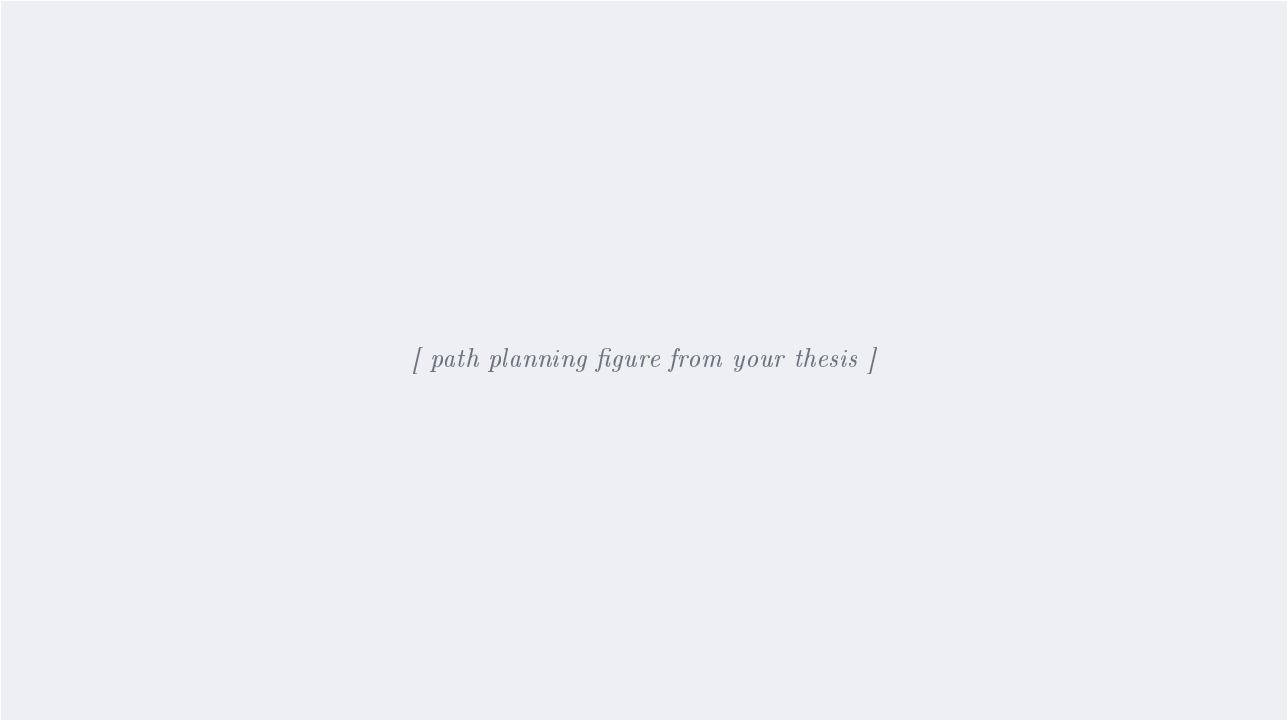
A set of simulations covering the main manipulator types — Cartesian, cylindrical, spherical, articulated, SCARA, and a 6-DOF SCARA V3 — built to test and teach kinematics, dynamics and control.

MY ROLE

Wrote all the simulations. They are used in the robotics courses I teach to third-year Mechatronics Engineering students.

TOOLS

MATLAB · Robotics Toolbox (Peter Corke)



[path planning figure from your thesis]

WHAT IT IS

My MS thesis: motion planning for a 6-DOF manipulator using neural eikonal fields. The method computes a cost field that the arm's path follows to reach a target while avoiding obstacles.

MY ROLE

[one or two lines — what you built and what results you have so far]

TOOLS

[your stack — e.g. Python, PyTorch, MATLAB]

Alongside the engineering, I make the material that explains it. I script, film and edit technical videos, write instruction materials, and keep public repositories that other people learn from.

[two or three YouTube thumbnails or a channel screenshot]

YOUTUBE — MECHATRONICS AND ROBOTICS ENGINEERING

Robotics 1 and 2, data science in Excel, Git and GitHub, basic workshop and machining. Written, filmed and edited by me.

GITHUB REPOSITORIES USED IN MY LECTURES

Robotics_MEXE_3rdYearCourse — kinematics, control algorithms, simulation tasks.
MexEE402_AI — data science and machine learning, model productionization.

PRINT DOCUMENTATION

Brochure and instruction materials for the solar power station.

Thank you for reading.

Mikko De Torres

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GitHub: [link]